Lecture 18

3/2/26
JRA

HK Finish std deblurring
Move to spectrum of special techniques
MT ≈ 2 wks

HW - pinhole confocal

LS pinhole in int
SD illum lots of pinholes at one time Fast Less process

Computational method Comp Deblurring / Deconv

Building up a 3D image from lesser info is pervasive

Want to know micros med 3D distribution of stuff in object

Microscope Get 3D optical sections
Distorted by diffraction
Get image of point source of light

→ bigger, dimmer

Imaging theory says

$$I = O \otimes PSF$$

3 things are related know 2

Convolution (an integral)

Know I & PSF
Solve for O

If $\otimes$ were simple mult could solve for O by dividing

Reality: ~~More complicated~~ Iteration

Start w/ "intelligent guess" for O
Best guess is $I = O$

Start w/ guess
Computer convolves guess w/ PSF
If guess is good

$$G \otimes PSF = I$$

Compare convolution w/ image & revise guess

Ex 50 / 100 raise by ratio

Iterate ≈ 20 times get

Guess too high

$G \times PSF = 200 \quad I = 100$

Lower G by $100/200$

~20 Iterations

Reconstruct object

Software approach complements HW approach
Can get 0

Often done in conjunction w/ confocal

Detour

* Polarization µscopy

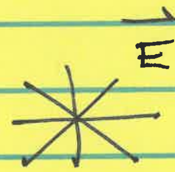
Look at ~~one~~ more very powerful method of looking at unstained samples

Light is composed of E & B fields

Natural light is coming from many atoms. Each atom light is polarized but sum is unpolarized

Natural light is "randomly" polarized

Each atom will generate light w/ a given E direction (electric field)

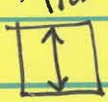


lots of atoms randomly polarized

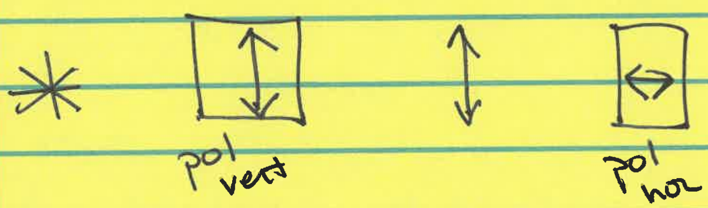
$\boxed{\text{DEMO}}$

One polarizer –
Two polarizers – crossed
Dark

Transmission axis



polarizer lets thru eg vertical light $\vec{E}$



Detector
No Light

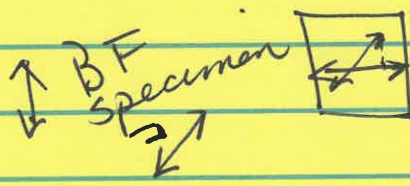
Crossed polarizers because transmission axes are crossed

Microscope can have two crossed polarizers
$\Rightarrow$ dark background

<u>Polarization μscope</u>

No specimen ⇒ no signal
dark background

Birefringent samples rotate light polarization



Some horizontal light will get thru

Specimen that is birefringent will create a signal

2) BF specimen will rotate light pol & create a ~~bright~~ signal due to lgth against a dark background

Things like μtubules are different ∥ & ⊥ to long axis. These work well w/ pol light

1950s Movie Meiosis

No staining

HW

Rules of the game w/ polarizers

(1) Unpolarized light hits a polarizer
Output intensity is down by a factor of two $\quad I_{out} = I_{in}/2$

& light is polarized along transmission axis

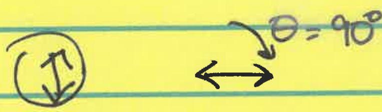
(2) Polarized light hits a polarizer

Malus's Law

pol. along trans axis of second pol

$$I_{out} = I_{in}\cos^2\theta$$

Polarizer w/ cocked transmission axis

Ex 

Crossed pol = 90°

$\boxed{\text{ACTIVITY 7}}$ Polarizers

1 Vert
2 60°
3 Hor

(a) A $\quad I_o/2 \quad$ vert

B $\quad I_o/8 \quad$ 60°

$\frac{I_o}{8}\cos^2 30°$ $\quad$ 30° $\quad$ C $\quad 3I_o/32 = 0.094\ I_o$ $\quad$ this is the signal caused by a polarizing specimen that rotates light

(b) A $\quad I_o/2$) crossed pol
C $\quad 0$ $\quad$ No signal

Reviewed answers

Sample (a) is like having a polarizing specimen

(B) 90° difference

$\boxed{\text{DEMO 2}}$ Add X polarizer